

PARAG DESHPANDE

Bengaluru, Karnataka, India

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Summary

Agentic AI Developer with **1.5+ years** designing and deploying **multi-agent systems, LLM-driven pipelines, and autonomous AI solutions** in enterprise environments. Hands-on with **Python, LangChain, CrewAI, RAG, Agentic Orchestration, and Vector Databases** – building agents capable of reasoning, planning, and executing tasks end-to-end. Proficient in **LLM Integration & Deployment, FastAPI, REST APIs, CI/CD, and cloud platforms (Azure/AWS)** with full-lifecycle ownership from POC to production deployment.

Education

MIT Institute of Technology

B.Tech in Computer Science and Engineering

2020 – 2024

Aurangabad, India

Experience

Software Engineer – AI & Data Solutions

Jan 2025 – Present

Tata Consultancy Services (TCS)

Bengaluru, India

- Designed and deployed **LangChain multi-agent pipelines** for enterprise document understanding – implementing **autonomous tool-use, chain-of-thought reasoning, context windowing, and JSON schema enforcement** for structured output generation.
- Implemented **RAG pipelines** with **ChromaDB vector database** for semantic retrieval – covering document ingestion, chunking, embedding, and context-aware response generation at enterprise scale.
- Built **modular agent tools and REST APIs** using **FastAPI** – enabling agents to autonomously interact with enterprise backends, databases, and microservices in production workflows.
- Owned **end-to-end LLM integration and deployment** – from model selection and prompt design to production rollout, monitoring, and evaluation across agent components.
- Integrated **automated evaluation pipelines** across agent components – validating retrieval accuracy, output consistency, and failure-mode behaviour against ground truth before enterprise rollout.
- Collaborated with **cross-functional teams** (product, QA, infra, domain experts) in **Agile/Scrum** – conducting POCs, model comparisons, and rapid iterations to optimize agent performance and reliability.
- 2nd Place, TCS AI Friday S2 Hackathon:** Built a **RAG-powered autonomous IT Helpdesk Assistant** (FastAPI + Groq/LLaMA + ChromaDB) – stakeholder-approved for **internal enterprise rollout**.

Digital Solutions Intern

Jan 2024 – Jul 2024

Expert Global Solutions

Aurangabad, India

- Developed **Python backend services** for analytics workflows using **SQL and MongoDB** – contributed to API integration, defect resolution, and Agile delivery.
- Built and optimized **Azure cloud-based pipelines** for high-volume data processing – ensuring scalability, fault tolerance, and reliability under production load.

Projects

LeadCommand – Scalable Multi-Agent AI Platform

2025 – Present

Python, FastAPI, CrewAI, LangChain, Groq/LLaMA, Supabase, Next.js, GitHub Actions

- Architected a **production multi-agent AI platform** using **CrewAI** – designing **autonomous agents with tool-use, memory, and agentic orchestration** for real estate ad automation end-to-end.
- Built **modular, independently deployable backend services** with FastAPI and CI/CD (GitHub Actions) – applying best practices in **guardrails, error handling, observability, and REST API design**.

RAG-Powered IT Helpdesk – Agentic NLP System

2025

Python, FastAPI, ChromaDB, Groq/LLaMA, LangChain, TF-IDF, Streamlit

- Built an **autonomous agent** combining **semantic RAG retrieval** (ChromaDB vector database) with an **NLP classifier** – implemented **fallback mechanisms and evaluation pipelines** to validate reliability before enterprise rollout.

Technical Skills

Programming: Python, SQL, Java, Pandas, NumPy

Agentic AI & Orchestration: LangChain, CrewAI, LangGraph (fundamentals), AutoGen (fundamentals), Multi-Agent Systems, Agentic Orchestration, Prompt Engineering

LLMs & Integration: LLM Integration & Deployment, RAG, LLaMA, GPT-4o, DeepSeek

Vector Databases & Memory: ChromaDB, FAISS (fundamentals), Pinecone (fundamentals), Semantic Search, Embedding Pipelines

Backend & Deployment: FastAPI, REST APIs, Microservices, CI/CD (GitHub Actions), Docker (fundamentals), Azure, AWS

ML & NLP: Scikit-learn, XGBoost, NLP, Feature Engineering, Model Evaluation, Statistical Analysis

Databases: PostgreSQL, MySQL, MongoDB, Supabase, ChromaDB

Certifications

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

Problem Solving (Intermediate) – HackerRank